Literature Review: Multi-Agent Rainwater Management for Urban Flood Mitigation

1. Introduction

This review asks what existing research offers for a team of classical-AI agents that must reroute and store rainwater during a storm, when sensors are noisy and drainage connections close unexpectedly. The project is a simulated multi-agent system for a synthetic Jeddah-like district: a graph of ten named zones — main roads, residential, industrial, school and coastal-park areas — linked by drainage connections, with two terminal storage basins, each zone having a fixed capacity and a defined drainage direction. Five agents coordinate through one shared environment model: Rainfall Monitoring (probabilistic updating of noisy sensor readings), Water Collection (rule-based decisions on where excess water is directed), Route/Drainage (A* routes around blocked connections, replanning when one closes), Storage Management (rule-based thresholds that reroute inflow from a nearly full basin to the basin with the most free capacity) and Emergency Response (rule-based prioritisation of the highest-risk zones). No machine learning is used. Success is measured against a fixed-drainage baseline by number of flooded zones, runoff stored and replanning success after injected faults.

The review covers fifteen peer-reviewed conference and journal papers. They are grouped by approach: flood-management analysis and decision support [1]–[5]; search-based path planning [6]–[8]; dynamic hazard-aware routing [9]; Bayesian fusion of rainfall sensor data [10]–[12]; multi-agent coordination through shared state [13]; and constrained water-resource allocation [14], [15]. Most records in the original export set were off-topic (consensus control, power grids, network routing), so targeted searches were run to add work on probabilistic rainfall sensing, hazard-aware A* routing and reservoir allocation, which the first draft of this review lacked.

2. Existing AI Approaches and Methodologies

Theme A: flood analysis and decision support. The flood-specific papers analyse systems rather than control them. Piro et al. survey real-time control of stormwater drainage, predictive models, decision support and early-warning systems [1]. Kamashki and Hdeib run hydrological and hydraulic assessments of the Al Lawzi drainage system in Bahrain under extreme-rainfall scenarios [2], and Sonavane et al. simulate Pune's drainage in EPA SWMM, finding surcharge at four junctions and recommending larger drains [3]. Hdeib ranks nature-based solutions for arid cities with multi-criteria decision analysis and finds rainwater harvesting best [4]. Ramachandran et al. use a random forest to site storm drains [5]; it is machine learning, included only as a contrast. All five treat the storm as a fixed input and produce a static design or ranking.

Theme B: search-based path planning. Ali and Das implement an A* variant on a 15×15 grid with velocity-obstacle avoidance [6], Yu et al. combine improved Theta* with B-spline smoothing for a vehicle [7], and Scheffe et al. recast receding-horizon control as graph search over a finite-state automaton, proving recursive feasibility with two vehicles in real time [8]. Together they show that grid search and horizon-limited planning are real-time capable, which supports the Route/Drainage and

Storage Management agents; horizon-based storage planning as in [8] is a possible later extension, not part of the current design. Every obstacle is, however, a physical object and not a hazard with a cost.

Theme C: dynamic hazard-aware routing. Priority-based Discrete-time Dynamic Routing (PDDR) models multi-source, multi-destination evacuation as network flow that prioritises high-risk locations and replans as the hazard changes; it beats a no-priority variant and static greedy routing [9]. This is the closest analogue to the Route/Drainage agent, but it addresses building evacuation, not rainwater, and assumes the hazard state is known.

Theme D: Bayesian fusion of rainfall sensor data. Yang and Ng propose a fast Bayesian regression-kriging method merging radar, rain-gauge and crowdsourced rainfall data in real time for urban storm applications [10]. Scheidegger and Rieckermann assimilate rainfall sensors with fundamentally different integration characteristics — gauges, radar and microwave links — over a small urban area using Bayesian assimilation [11]. Hu et al. review rainfall spatial estimation and multi-source data merging, covering Bayesian filtering among the assimilation methods [12]. Together these establish Bayesian updating as the standard technique for reconciling noisy, heterogeneous rainfall observations in real time. None feeds the posterior estimate to an agent that acts on it: the output is a corrected rainfall field, not a routing or storage decision.

Theme E: coordination and constrained allocation. Yang and Liu coordinate satellite agents through a blackboard with serial or parallel worker activation, reporting mean fitness up to 20% higher than the best centralised algorithm and negotiation more than 80% cheaper than contract-net [13]; the problem solver is a hybrid metaheuristic, not classical search, but the shared-state coordination supports the project's environment model. Sawangphol et al. formulate reservoir operation as a constraint programming problem, allocating limited water among distribution zones [14]. Yang et al. address rainstorm-induced disruption directly, allocating mobile pumping resources across a multimodal transport network using rolling optimization guided by predicted flood growth [15]. Both treat allocation as offline or rolling optimization over a fixed network and neither couples allocation to real-time hazard-aware routing by cooperating agents.

3. Critical Comparison

The studies split on one axis: those closest to the flood problem use no classical search or reasoning, and those that use search, Bayesian inference or constraint programming never manage a storm event through cooperating agents. Table I summarises this. Results are taken from abstracts; where an abstract reports no figure the cell says so, and entries marked [VERIFY] require confirmation against the full text.

Ref	AI Technique	Problem / Test Cases	Evaluation & Result	Limitation
[1]	Real-time control, predictive models, decision support (survey)	Urban stormwater drainage and early warning	Qualitative review; no test cases [VERIFY]	No implementation or metrics
[2]	Hydrological/hydraulic modelling	Al Lawzi lake system, Bahrain; extreme-rainfall	Capacity weaknesses identified; no figures in abstract	Static design analysis; no runtime adaptation

Ref	AI Technique	Problem / Test Cases	Evaluation & Result	Limitation
		scenarios		
[3]	EPA SWMM simulation with GIS	Pune drainage; past critical storms	Surcharge at J-3, J-9, J-11, J-21; larger drains advised	Pre-2021; fixed events; no agents
[4]	Multi-criteria decision analysis	Nature-based options for arid cities	Rainwater harvesting ranked first; score not in abstract	Weights and data [VERIFY]; static ranking
[5]	Random forest (ML, out of scope)	Storm-drain siting by district	Parameter influence analysed; no accuracy in abstract	ML; static siting; official data scarce
[6]	A* grid search; velocity-obstacle avoidance	15×15 grid, moving obstacles	Reported efficient and reliable; no numbers in abstract	Small grid; no hazard costs; no uncertainty
[7]	Improved Theta* with B-spline	Vehicle; two static/moving-obstacle scenarios	Effective in simulation; no numbers in abstract	Kinematics-specific; two scenarios; no baseline
[8]	Graph search over finite-state automaton; receding horizon	Two-vehicle trajectory planning, lab and simulation	Real-time; recursive feasibility proved	Only two agents; no uncertainty
[9]	Network-flow dynamic routing with priorities	Multi-source evacuation in closed spaces	PDDR beats DDR and SGR; margins not in abstract	Buildings, not floods; hazard assumed known
[10]	Bayesian regression kriging (sensor fusion)	Distributed Radar + gauge + crowdsourced rainfall, urban stormmission planning	Fitness up to 20% above centralised; Real-time merging of heterogeneous sensors [VERIFY metrics]down more than 80%	Satellites; offline; Produces a rainfall field; no downstream agentclassical search
[11]	Bayesian assimilation of heterogeneous sensors	Urban area; gauges, radar, microwave links	Assimilation demonstrated across sensor types	Pre-2021; no control or routing loop
[12]	Review of spatial interpolation and multi-source	Rainfall estimation methods broadly	Surveys Bayesian filtering among assimilation methods	Review only; no agent design

Ref	AI Technique	Problem / Test Cases	Evaluation & Result	Limitation
	merging			

Table I. Summary of reviewed studies.

Three points emerge. First, evaluation is weak where relevance is highest: the flood papers [1]–[5] compare against no baseline, while only [9], [13] and [15] report against an alternative, and [6]–[8] validate in simulation alone. Second, uncertainty and action are separated: [10]–[12] handle noisy sensing but stop at the estimate, while [9] and [15] act but assume the hazard state is known. Third, routing and storage are never coupled: the planners [6]–[9] avoid obstacles without allocating a limited resource, and the allocators [14], [15] manage water or pumps without hazard-aware pathfinding by cooperating agen**ts.**

4. Limitations and Research Gap

No reviewed paper combines Bayesian sensing under noise, hazard-aware search and resource allocation during the storm itself, coordinated by multiple agents. The gaps are:

1. Static versus in-event decisions. Flood studies [1]–[5] size or rank infrastructure for a design storm; none replans as the event unfolds.

2. Sensing without action. Bayesian fusion [10]–[12] produces a corrected rainfall estimate but never feeds it to an agent that reroutes or stores water.

3. Action without uncertainty. Hazard-aware routing [9] and resource scheduling [15] assume the hazard state is observed, not inferred from noisy sensors.

4. Routing without storage, storage without routing. A* and graph-search planners [6]–[9] treat blocked cells as binary and ignore limited storage; constraint and optimization allocators [14], [15] allocate capacity without hazard-aware pathfinding.

5. Weak evaluation. Few studies state a baseline, run counts or fault injection, so claims such as "effective" or "outperforms" cannot be compared.

The project addresses four of these five gaps and leaves one open. Rainfall Monitoring infers rainfall from noisy readings by probabilistic updating and shares it with the other agents (gap 2). Route/Drainage replans A* routes around closed drainage connections, but treats a closure as binary and does not weight routes by flood risk, so gap 3 remains open. Storage Management reacts to basin capacity through thresholds, a first step toward gap 4; horizon-based reservoir allocation is left for later iterations. The agents coordinate through one shared environment model, in the spirit of the shared-state coordination in [13]. Evaluation (gap 5) is addressed by comparison with a fixed-drainage baseline over 30 runs per

scenario at 10, 30 and 60 mm/h, and under 20% sensor dropout, Gaussian noise (σ = 5 mm/h) and one to three drainage-connection closures.

5. References

[1] P. Piro, M. M. Saleh, B. Pirouz, M. Turco, and S. A. Palermo, "Smart and innovative systems for urban flooding risk management," in *Proc. 2023 Int. Conf. Inf. Commun. Technol. Disaster Manage. (ICT-DM)*, 2023, pp. 1–4, doi: 10.1109/ICT-DM58371.2023.10286951.

[2] R. Kamashki and R. Hdeib, "Towards resilient stormwater management system. Application of AI-Lawzi Lake stormwater system," in *Proc. 2024 ASU Int. Conf. Emerg. Technol. Sustain. Intell. Syst. (ICETSIS)*, 2024, pp. 182–187, doi: 10.1109/ICETSIS61505.2024.10459470.

[3] N. Sonavane, V. A. Rangari, M. L. Waikar, and M. Patil, "Urban storm-water modeling using EPA SWMM — a case study of Pune city," in *Proc. 2020 IEEE Bangalore Humanitarian Technol. Conf. (B-HTC)*, 2020, pp. 1–5, doi: 10.1109/B-HTC50970.2020.9297900.

[4] R. Hdeib, "Nature-based solutions for sustainable urban drainage: Implications for arid region urban resilience," in *Proc. 2025 Int. Conf. Decis. Aid Sci. Appl. (DASA)*, 2025, pp. 2337–2343, doi: 10.1109/DASA68193.2025.11499033.

[5] R. Ramachandran et al., "Harnessing QGIS and random forest analysis for precision storm drain reinforcement," in *Proc. 2024 Int. Conf. Emerg. Res. Comput. Sci. (ICERCS)*, 2024, pp. 1–5, doi: 10.1109/ICERCS63125.2024.10895146. [VERIFY full author list — export gives initials only]

[6] M. Ali and S. Das, "Mobile robots with dynamic obstacle avoidance," in *Proc. 2023 IEEE 3rd Int. Conf. Sustain. Energy Future Elect. Transp. (SEFET)*, 2023, pp. 1–6, doi: 10.1109/SeFeT57834.2023.10245465.

[7] H. Yu, X. Wang, and W. Sun, "An improved Theta*-based trajectory planner for autonomous vehicle with obstacle avoidance," in *Proc. 2021 4th IEEE Int. Conf. Ind. Cyber-Phys. Syst. (ICPS)*, 2021, pp. 613–618, doi: 10.1109/ICPS49255.2021.9468211.

[8] P. Scheffe, M. V. A. Pedrosa, K. Flaßkamp, and B. Alrifaee, "Receding horizon control using graph search for multi-agent trajectory planning," *IEEE Trans. Control Syst. Technol.*, vol. 31, no. 3, pp. 1092–1105, May 2023, doi: 10.1109/TCST.2022.3214718.

[9] S. N. P, J. Bapat, and D. Das, "Priority based discrete time dynamic routing algorithm for emergency evacuation," in *Proc. 2021 IEEE Int. Conf. Electron., Comput. Commun. Technol. (CONECCT)*, 2021, pp. 1–6, doi: 10.1109/CONECCT52877.2021.9622540. [VERIFY first author's full name]

[10] P. Yang and T. L. Ng, "Fast Bayesian regression kriging method for real-time merging of radar, rain gauge, and crowdsourced rainfall data," *Water Resources Research*, vol. 55, 2019. [VERIFY volume/issue/pages and DOI]

[11] A. Scheidegger and J. Rieckermann, "Bayesian assimilation of rainfall sensors with fundamentally different integration characteristics," in *Proc. Int. Symp. Weather Radar and Hydrology*, 2014. [VERIFY venue, pages, DOI]

[12] Q. Hu, Z. Li, L. Wang, Y. Huang, Y. Wang, and L. Li, "Rainfall spatial estimations: A review from spatial interpolation to multi-source data merging," *Water*, vol. 11, no. 3, 2019. [VERIFY article number and DOI]

[13] Y. Yang and D. Liu, "Distributed imaging satellite mission planning based on multi-agent," *IEEE Access*, vol. 11, pp. 65530–65545, 2023, doi: 10.1109/ACCESS.2023.3289964.

[14] W. Sawangphol, J. Kraisangka, A. Rittima et al., "Constraint programming for reservoir operation optimization of Bhumibol dam," *Applied Water Science*, 2024. [VERIFY volume, pages, DOI]

[15] H. Yang, R. Zhang, W. Zhou, and L. Hui, "A graph-spatiotemporal prediction-aware optimization framework for emergency resource scheduling in rainstorm-induced multimodal transportation networks," *IEEE Access*, vol. 14, 2026. [VERIFY pages, DOI]

Conference locations are omitted because the exports do not include them.

Appendix: Verification Log

The title, venue, year and DOI below were checked against the IEEE Xplore exports you supplied. I could not open the full papers in this environment, so the last column is for the team to complete after opening each PDF; the results in Table I come from abstracts only.

Ref	Title	Venue	Year	DOI	Checked against export	Full text opened by team
[1]	Smart and Innovative Systems for Urban Flooding Risk Management	ICT-DM	2023	10.1109/ICT-DM58371.2023.10286951	Yes	[ ]
[2]	Towards Resilient Stormwater Management System	ICETSIS	2024	10.1109/ICETSIS61505.2024.10459470	Yes	[ ]
[3]	Urban storm-water modeling using EPA SWMM	B-HTC	2020	10.1109/B-HTC50970.2020.9297900	Yes	[ ]
[4]	Nature-Based Solutions for Sustainable Urban Drainage	DASA	2025	10.1109/DASA68193.2025.11499033	Yes; export's abstract and IEEE Terms are garbled	[ ]
[5]	Harnessing QGIS and Random Forest Analysis	ICERCS	2024	10.1109/ICERCS63125.2024.10895146	Yes	[ ]
[6]	Mobile Robots with Dynamic Obstacle Avoidance	SEFET	2023	10.1109/SeFeT57834.2023.10245465	Yes	[ ]
[7]	An Improved Theta*-based	ICPS	2021	10.1109/ICPS492	Yes	[ ]

Ref	Title	Venue	Year	DOI	Checked against export	Full text opened by team
	Trajectory Planner			55.2021.9468211		
[8]	Receding Horizon Control Using Graph Search	IEEE Trans. Control Syst. Technol.	2023	10.1109/TCST.2022.3214718	Yes	[ ]
[9]	Priority based Discrete time Dynamic Routing Algorithm	CONECCT	2021	10.1109/CONECCT52877.2021.9622540	Yes	[ ]
[10]	Distributed Imaging Satellite Mission Planning Based on Multi-Agent	IEEE Access	2023	10.1109/ACCESS.2023.3289964	Yes	[ ]
[11]	Fast Bayesian Regression Kriging	Water Resources Research	2019	[VERIFY]	Partial — abstract only	[]
[12]	Bayesian Assimilation of Rainfall Sensors	Weather Radar & Hydrology	2014	[VERIFY]	Partial — abstract only	[]
[13]	Rainfall spatial estimations: A review	Water	2019	[VERIFY]	Partial — abstract only	[]
[14]	Constraint Programming for Reservoir Operation	Applied Water Science	2024	[VERIFY]	Partial — abstract only	[]
[15]	Graph-Spatiotemporal Prediction-Aware Optimization	IEEE Access	2026	[VERIFY]	Partial — abstract only	[]